



檢驗醫學

Laboratory
Medicine

LABORATORY MEDICINE
LABORATORY MEDICINE
LABORATORY MEDICINE
LABORATORY MEDICINE

學號：

姓名：

Vol.

1

BLOCK 考古題目

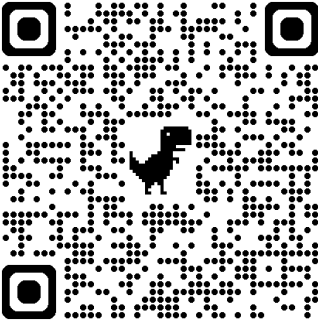
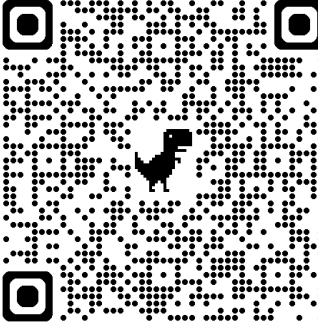
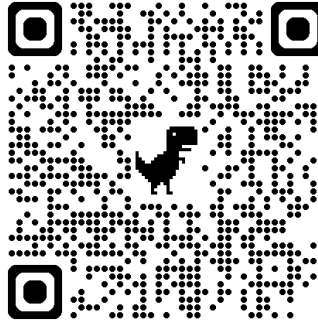
考古製作群		
寫手： 黃芊裴 黃謙愛 劉詠欣 黃芷宣	組長： 林煜翔	共筆長： 郭祐維 王泓予 康元輔 蕭婷云

重要公布事項：

寫手想跟你 / 妳各位說：

組長想跟你 / 妳各位說：

共筆長想跟你 / 妳各位說：

勘誤表單：	勘誤表	想說的話
		

--

目錄

B99~B03 分類考古	1
§ 鐘桂彬老師—檢驗結果的判讀與品質	1
§ 鄭文誠老師—重點照護檢驗	1
§ 鄭文誠老師—藥物濃度監測	3
§ 詹一秀老師—體腔液檢驗	5
§ 分子檢驗學	8
 B04~B10 歷屆考古	 12
§ B04 第一次區段考	12
§ B05 第一次區段考	14
§ B06 第一次區段考	16
§ B07 第一次區段考	20
§ B08 第一次區段考	23
§ B09 第一次區段考	25
§ B10 第一次區段考	28

B99~B03 分類考古

§ 鐘桂彬老師—檢驗結果的判讀與品質

1. 請問如右圖示代表實驗數據的可信度為何？（B03, 02）

- (A). Accurate and precise
- (B). Not accurate, but precise
- (C). Accurate, but not precise
- (D). Neither accurate nor precise



2. 某檢測的偵測結果如下表：（B03, 02）

	Disease	No disease
Positive	196	20
Negative	4	180

- (1) 請問其 Specificity（特異度）為多少？
- (2) Negative predictive value 為多少？

§ 鄭文誠老師—重點照護檢驗

1. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in junior emergency medicine unit (as below). According to the results, which one is incorrect? (B03)

	Geometric mean (hours)		% Reduction (95% CI)	p Value
	Control	POC		
Overall	3.50	3.24	7.6 (0.4 to 14.3)	0.04
Diagnostic group				0.86*
ACS	3.43	3.15	8.2 (-0.9 to 16.5)	0.08
General	3.61	3.36	6.9 (-5.2 to 17.6)	0.25
Disposition				0.60*
Discharge home	3.68	3.50	4.9 (-5.9 to 14.5)	0.36
Admit to ward	3.66	3.22	12.1 (-0.8 to 23.3)	0.06
Admit to EMU	2.94	2.81	4.4 (-11.1 to 17.8)	0.56
Enrolling staff				0.06*
Consultant or registrar	3.63	2.94	19.1 (7.3 to 29.4)	<0.01
Junior medical officer	3.54	3.51	0.9 (-9.6 to 10.4)	0.85
Nurse	3.24	3.12	3.8 (-15.7 to 20.1)	0.68

The bold text represent the results for the whole group.
 *Testing for an interaction between intervention and subgroup.
 ACS, acute coronary syndrome; EMU, Emergency Medicine Unit; POC, point-of-care.

- (A). POCT may shorten the disposition time overall.
- (B). Senior staffs (consultant or registrar) may shorten disposition time by POCT.
- (C). Junior medical officer may not shorten disposition time by POCT.
- (D). Nurses may shorten disposition time by POCT according to this study.

2. About point of care testing (POCT) which is wrong? (B03)
- (A). Shorten the therapeutic turnaround time of laboratory tests.
 - (B). Reduces length of hospital stay due to faster laboratory services.
 - (C). Only can be applied within hospitals.
 - (D). More expensive than those of central laboratory testing.
3. Advantages of point-of-care testing (POCT), which is wrong? (B02)
- (A). Shorten the therapeutic turnaround time.
 - (B). Reduces clinical visits, hospital admission, length of hospital stay due to faster laboratory services.
 - (C). services.
 - (D). More expensive than those of central laboratory testing.
 - (E). Availability to a wider variety of sites (e.g., rural areas, areas with limited infrastructure/personal).
4. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in junior emergency medicine unit (as below). According to the results, which one is incorrect? (B02, 01 ; 圖見第一題)
- (A). POCT may shorten the disposition time overall.
 - (B). Senior staffs (consultant or registrar) may shorten disposition time by POCT.
 - (C). Junior medical officer may not shorten disposition time by POCT.
 - (D). POCT is both suitable in ACS patients and general (non-ACS) ones according to this study.
5. Disadvantages of point of care testing (POCT), which is wrong? (B01)
- (A). Does not shorten the therapeutic turnaround time.
 - (B). Increasing the cost per test.
 - (C). Difficult to manage quality control/assurance and maintenance.
 - (D). Heavy billing and regulatory restriction recently.
6. Advantages of point of care testing (POCT), which is/are correct? (B01, 00)
- (A). Shortening the therapeutic turn-around time.
 - (B). Decreasing the cost per test.
 - (C). Easier to manage quality control/assurance and maintenance.
 - (D). Less billing and regulatory restriction recently.

§ 鄭文誠老師—藥物濃度監測

1. Pharmacokinetics includes:

- (1) Drug absorption
- (2) Drug distribution
- (3) Drug-drug interaction in metabolism
- (4) Drug action at target site
- (5) Drug excretion

Which is incorrect? (B03)

- (A).12345
- (B).4
- (C).34
- (D).234

2. Clinically utility of therapeutic drug monitoring (TDM) includes:

- (1) compliance of patients with epilepsy
- (2) patients with epilepsy in pregnancy
- (3) adjust the dosage of tacrolimus in patients' status post renal transplantation
- (4) adjust the dosage of amikacin in patients with septicemia
- (5) adjust the dosage of rituximab in patients with diffuse large B cell lymphoma.

Which one(s) is (are) incorrect? (B03)

- (A).15
- (B).5
- (C).35
- (D).34
- (E).3

3. Pharmacokinetics includes:

- (1) Drug absorption
- (2) Drug distribution
- (3) Drug-drug interaction and metabolism
- (4) Drug action at target site
- (5) Drug excretion

Which is correct? (B02)

- (A).12345
- (B).1235
- (C).1345
- (D).2345

4. Clinically utility of therapeutic drug monitoring (TDM) includes:

- (1) compliance of patients with epilepsy
- (2) patients with epilepsy on pregnancy
- (3) adjust the dosage of tacrolimus in patients' status post renal transplantation
- (4) adjust the dosage of amikacin in patients with septicemia

Which one(s) is (are) wrong? (B02)

- (A).1
- (B).234
- (C).24
- (D).3

5. Pharmacokinetics include all the following except? (B101 - 99)

- (A). Drug absorption
- (B). Drug distribution
- (C). Drug interaction with target site
- (D). Drug metabolism

6. Clinically utility of therapeutic drug monitoring (TDM) includes all the following except?
(B101 - 99)

- (A). Compliance of patients with epilepsy
- (B). Patients with epilepsy in pregnancy
- (C). Adjust the dosage of Tacrolimus for patients' status post renal transplantation
- (D). Adjust dosage of metformin as hypoglycemic control in DM patient

7. For analytical aspects of cyclosporine A (CsA) in TDM, we should consider: (B00, 99 ; 多選)

- (A). Proficiency testing of CsA
- (B). Therapeutic and toxic level of CsA
- (C). Sampling time (through)
- (D). Methodology of CsA measurement

§ 詹一秀老師—體腔液檢驗

1. 下列何種病變會產生體腔滲出液 (Transudate) ? (B00)
 - (A). 系統性紅斑性狼瘡
 - (B). 腫瘤
 - (C). 腎病症候群
 - (D). 胰臟炎
2. 下列何者不是常規腹水檢查的項目 ? (B02)
 - (A). Complement
 - (B). Specific gravity
 - (C). Appearance
 - (D). Cell count
3. 下列何者不是常規腹水檢查的項目 ? (B00)
 - (A). Appearance
 - (B). BUN
 - (C). Liu's stain
 - (D). Protein
4. 下列何者不是常規心包膜積液檢查的項目 ? (B01)
 - (A). Appearance
 - (B). Liu's stain
 - (C). Amylase
 - (D). Total protein
5. 進行腹水細胞的分類時，下列何種方法最正確 ? (B99)
 - (A). 加熱固定染色法
 - (B). 劉氏染色法 (Liu's stain)
 - (C). 超活體染色法
 - (D). 革蘭氏染色法
6. 結核性肋膜炎的病患，胸水 (Pleural effusion) 中哪種白血球會增加 ? (B02, 01)
 - (A). 嗜中性白血球
 - (B). 嗜酸性白血球
 - (C). 淋巴球
 - (D). 中皮細胞
7. 結核性肋膜炎的病患，體腔液中哪種白血球會增加 ? (B00)
 - (A). 嗜中性白血球
 - (B). 嗜酸性白血球
 - (C). 嗜鹼性白血球
 - (D). 淋巴球
8. 藥物反應的病患或是淋巴瘤的病患，體腔液中哪種白血球會增加 ? (B99)
 - (A). 嗜中性白血球
 - (B). 嗜酸性白血球
 - (C). 嗜鹼性白血球
 - (D). 淋巴球
9. 下列何者是肋膜腔 (胸膜腔) 之滲 (漏) 液 (Transudate) 的特性 ? (B02)
 - (A). 肋膜腔積液 (胸膜腔積液，胸水) 的 Total protein 高於 50% 血清的 Total protein
 - (B). 肋膜腔積液 (胸膜腔積液，胸水) 的 LDH 低於 60% 血清的 LDH
 - (C). 白血球較多，通常 $> 1000/\text{mm}^3$
 - (D). 比重 > 1.018

10. 下列何者不是肋膜腔（胸膜腔）之滲（漏）液（Transudate）的特性？（B01）
- (A).肋膜腔積液（胸膜腔積液，胸水）的 Total protein 低於 50% 血清的 Total protein
 - (B).肋膜腔積液（胸膜腔積液，胸水）的 LDH 高於 60% 血清的 LDH
 - (C).白血球較少，通常 $< 1000 /\text{mm}^3$
 - (D).比重 < 1.015
11. 下列何者不是肋膜腔/胸膜腔之滲出液（Exudate）的特性？（B00）
- (A).會凝固
 - (B).肋膜腔積液（胸膜腔積液，胸水）的 LDH 低於 60% 血清的 LDH
 - (C).白血球較多，通常 $> 1000 /\text{mm}^3$
 - (D).比重 > 1.018
12. 下列關於體腔液檢查，檢體凝固現象的敘述，何者正確？（B98）
- (A).胸水凝固，代表採檢良好
 - (B).要加抗凝固劑至胸水檢體中，防止凝固，才容易進行細胞計數
 - (C).腹水不會凝固
 - (D).不要加抗凝血劑至腹水檢體中，使其凝固，才容易進行細胞計數
13. 下列何者為體腔的上皮細胞？（B99）
- (A).鱗狀細胞
 - (B).基底細胞
 - (C).內皮細胞
 - (D).中皮細胞
14. 下列何者不會造成微血管膠體滲透壓（Colloid osmotic pressure）減小，導致漿液（Serous effusion）形成增加？
- (A).肝硬化
 - (B).細菌性肺炎
 - (C).腎病症候群
 - (D).營養不良導致白蛋白（Albumin）減少
15. 何者 Rivalta 測試會呈陰性？
- (A).轉移癌胸水
 - (B).結核性胸水
 - (C).自來水
 - (D).胸水滲透液（Exudate）
16. 何者不是 Exudate 的特性？
- (A).比重 > 1.018
 - (B).會凝固
 - (C).肋膜 / 腔積液低於 50% Total protein
 - (D).肋膜 / 腔積液高於 60% LDH
17. 下列何種疾病不會造成微血管內膠體滲透壓（Colloid osmotic pressure）減少，導致漿液（Serous effusion）形成增加？
- (A).肺炎
 - (B).肝硬化
 - (C).腎病症候群
 - (D).營養不良導致白蛋白（Albumin）減少
18. 下列何者 Rivalta 試驗會呈現陽性反應？
- (A).心衰竭患者之胸水（Pleural effusion）
 - (B).自來水

- (C).肋膜腔 (Pleural cavity) 抽出之膿液
(D).冰醋酸
19. 下列何者不是胸水 (Pleural effusion) 之濾出液 (Transudate) 的特性？
(A).比重 ≤ 1.015
(B).因濾出液蛋白質濃度低，所以胸水檢體可以不加抗凝固劑
(C).胸水的 Total protein 低於 50% 血清的 Total protein
(D).胸水的 LDH 低於 60% 血清的 LDH
20. 下列有關體腔液之敘述，何者正確？
(A).濾出液 (Transudates) 常出現於肺炎
(B).滲出液 (Exudates) 常出現於腎病症候群
(C).濾出液常出現於肝硬化
(D).濾出液 (Transudates) 常出現於癌症轉移
21. 檢查血清 (serum) 與腹水蛋白質濃度差異之目的為何？
(A).判斷是否正確採檢
(B).判斷檢體是否新鮮
(C).判斷病人是否空腹
(D).判斷腹水為濾出液或滲出液
22. 下列何者不是造成胸膜腔 (pleural cavity) 積水的原因？
(A).微血管膠體滲透壓 (colloid osmotic pressure) 減少
(B).淋巴管再吸收減少
(C).微血管通透性 (permeability) 減少
(D).微血管液體靜水壓力增加 (hydrostatic pressure)
23. 下列何者 Rivalta 試驗會呈現陰性反應？
(A).轉移癌胸膜液
(B).滲出液
(C).濾出液
(D).結核性胸膜液 (tuberculous pleural effusion)
24. 下列何種病變會產生滲出液 (Exudate)？(B02 ~ 99，多選)
(A).心衰竭
(B).鬱血性心衰竭 (CHF)
(C).心律不整
(D).肝衰竭
(E).肝硬化
(F).結核性肋膜炎
(G).細菌性肋膜炎
(H).肺炎併肋膜腔積液
(I).系統性紅斑性狼瘡 (SLE) 併胸膜炎
(J).腎病症候群

§ 分子檢驗學

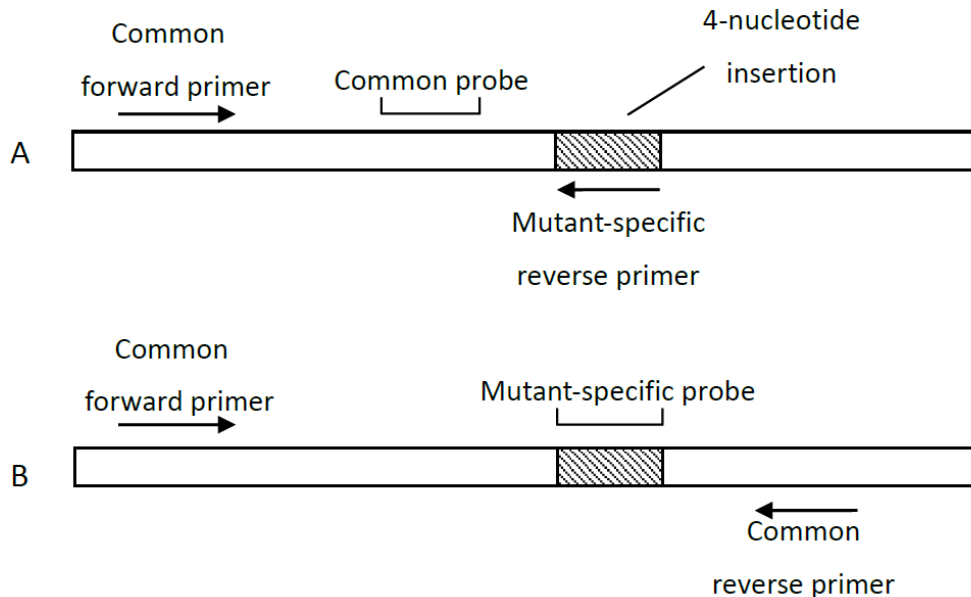
1. Which of the following translocation leads to a fusion protein? (B01)
 - (A).t (9; 22) (q34; q11.2)
 - (B).t (8; 14) (q24; q32)
 - (C).t (2; 8) (p12; q24)
 - (D).t (8; 22) (q24; q11)
 - (E).t (11; 14) (q13; q32)
2. Which of the following translocation implies high sensitivity to imatinib? (B03, 01)
 - (A).t (14; 18) (q32; q21)
 - (B).t (9; 22) (q34; q11.2)
 - (C).t (8; 21) (q24; q11)
 - (D).t (8; 14) (q24; q32)
 - (E).t (15; 17) (q22; q12)
3. JAK V617F mutation is frequently seen in which of the following diseases? (B01)
 - (A).Myeloproliferative neoplasm
 - (B).Myelodysplastic syndrome
 - (C).Acute lymphoblastic leukemia
 - (D).Chronic myeloid leukemia
 - (E).Hodgkin's lymphoma
4. Which of the following statement is correct? (B03, 01)
 - (A).Promotor sequence is a part of exons of a gene.
 - (B).5' untranslated region is a part of exons of a gene.
 - (C).Transcription of a gene terminates at TGA, TAA or TAG.
 - (D).The start codon of a protein is always in the first exon.
 - (E).The stop codon of a protein is always in the last exon.
5. Which of the following techniques is best suitable for detection of large insertion (e.g. Up to 5 kb) in a specific area of genome? (B98, 97)
 - (A).PCR
 - (B).Northern blot
 - (C).Southern blot
 - (D).Next generation sequencing
 - (E).Pyrosequencing
6. Which of the following description is right? (B98)
 - (A).Insertion or deletion mutations in a gene always disrupt the reading frame.
 - (B).The lengths of the coding sequence and the mRNA transcript of a gene are usually equal.
 - (C).The mRNA sequence of a gene always contains nucleotides whose number is multiple of 3.
 - (D).In human, there are total 61 possible combinations of amino acid codons (excluding stop codon).
 - (E).In human, every amino acid is encoded by more than 1 codon.
7. In which of the following situations is in vitro PCR advantageous when compared with DNA amplification by bacterial cloning? (B98)
 - (A).Large fragment of DNA (e.g. 10000 bp) is to be amplified.
 - (B).Amplification needs to be completed in 1 day.
 - (C).Large amount of products are desired.
 - (D).Mutations during amplification are not allowed.
 - (E).GC-rich sequences are frequent in the amplified product.
8. Which of the following conditions may imply that a nucleotide variant seen in cancer cells is likely a mutation with pathophysiological consequences? (B98)
 - (A).The variant is a silent mutation.
 - (B).The variant is a missense mutation.
 - (C).The variant is a nonsense mutation.
 - (D).The variant is a frameshift mutation.
 - (E).The variant is a deletion mutation.

- (A). The variant is also seen in the patient's non-cancerous tissue, such as buccal mucosa.
 - (B). The variant is within the middle of intron.
 - (C). The variant is within a string of nucleotides not well conserved between humans and mice.
 - (D). The variant is expected to disrupt the ATP binding to the kinase domain of a tyrosine kinase.
 - (E). The variant does not change the amino acid identity
9. Which of the following description is not appropriate for southern blot analysis? (B98)
- (A). Southern blot is usually used to detect point mutation in a gene.
 - (B). Southern blot usually involves restriction enzyme digestion.
 - (C). Southern blot utilizes capillary force to mobilize DNA.
 - (D). Southern blot involves hybridization between a probe and a target DNA.
 - (E). Southern blot needs immobilization of DNA on a membrane.
10. Which of the following protein products of chromosome translocation is not a fusion gene? (B97)
- (A). IgH-Myc
 - (B). CBFB-MYH11
 - (C). BCR-ABL
 - (D). AML1-ETO
 - (E). PML-RARA
11. Which of the following methods is not likely to increase the specificity and sensitivity of a PCR reaction? (B97)
- (A). Nested PCR by incorporation one pair of extra internal primers.
 - (B). Semi-nested PCR by incorporation one primer at either end.
 - (C). Repeated PCR using the same primer pair but with the product from the first round of PCR as a template.
 - (D). Adjust buffer and ion concentrations.
 - (E). Designing a new pair of primers with better annealing specificity.
12. Which of the following conditions may imply that a nucleotide variant is not likely a mutation with pathophysiological consequences? (B97)
- (A). The variant is not seen in 500 healthy controls.
 - (B). The variant is expected to cause protein truncation.
 - (C). The variant is well conserved among different species.
 - (D). The variant is within a middle part of an intron.
 - (E). The variant causes shifting of a reading frame, and the expected protein product is longer than the wild type.
13. Please choose 6 appropriate statements. (B00)
- (A). Usually the promotor regions are composed by exons.
 - (B). The 5' and 3' untranslated regions are composed by exons.
 - (C). There can be more than one protein products from a gene because of alternative splicing
 - (D). The number of nucleotide base pairs in coding sequences is a multiple of 3.
 - (E). Nest PCR can increase the specificity of a PCR reaction, reducing the production of unexpected sequence.
 - (F). Nest PCR can increase the specificity of a PCR reaction and also reduce the likelihood of cross contamination.
 - (G). When comparing the DNA amplification by PCR and by bacterial cloning, PCR usually produces more errors during amplification.
 - (H). t (8; 14) (q24; q32) results in translocation between c-MYC and IgH genes and leads to novel fusion protein.
 - (I). t (9; 22) (q34; q11) results in translocation between BCR and ABL1 genes and leads to novel fusion protein.

- (J). Whole exome sequencing is used to get sequencing of exons in the genome. It can be achieved by next genome sequencing, but this method usually is not able to detect fusion between two genes.
- (K). Real-time PCR is usually less sensitive than FISH (Fluorescence in situ hybridization) in minimal residual disease.
- (L). Capillary electrophoresis on fluorescence-tagged PCR products is frequently used to detect gene mutation. This method is especially sensitive in detecting point mutation.

14. Which approach will be more sensitive for minimal residual disease detection? Why?
(B02)

A 4-bp insertion mutation is known specific to a disease. Two students try to develop a real-time PCR-based quantitative approach to measure the mutation in patients' samples. Student A chooses method A (shown below as A), in which the reverse primer is specific to the insertion sequence, while the probe and forward primer bind non-mutation sequences. Student B chooses method B (shown below as B), in which the probe is mutation-specific, while the forward and reverse primers bind non-mutation sequences.



15. What is "Cycle of threshold" in real time PCR? What is its application in real time PCR?
(B99)
16. What is "Short tandem repeat" in human DNA? What is its application in biology?
(B99)

B04~B10 歷屆考古**§ B04 第一次區段考**

1. 檢驗人員打電話來通知說病患的檢體呈現溶血 3+，可能會導致血鉀濃度偽性偏高，建議重新抽血取得檢體。請問溶血對血鉀濃度所造成的誤差，屬於檢驗流程中哪一階段的問題？
 - (A). 檢驗前 (Pre-analytic)：造成抽血過程的困難
 - (B). 檢驗中 (Analytic)：檢驗方法學上面的干擾
 - (C). 檢驗後 (Post-analytic)：報告結果發送上的問題
 - (D). AB 皆對
 - (E). ABC 皆對
2. 以下何者為定量檢驗？
 - (1) 尿液蛋白 2+
 - (2) 血鉀濃度 75 mmol/L
 - (3) Tented T wave
 - (4) C 型肝炎檢驗陰性
 - (5) _____ 5.5 g/dL
 - (A). 12
 - (B). 23
 - (C). 25
 - (D). 14
 - (E). 45
3. About point of care testing (POCT), which is correct?
 - (1) Shorten the therapeutic turnaround time due to faster laboratory tests,
 - (2) reduces length of hospital stay due to faster laboratory services,
 - (3) only can be applied within hospitals, and
 - (4) more expensive than those of central laboratory testing.
 - (A). 12
 - (B). 13
 - (C). 23
 - (D). 124

4. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in Junior emergency medicine unit (as below). According to the results, which one is incorrect?

Table 2 Time from arrival to disposition decision

	Geometric mean (hours)		% Reduction (95% CI)	p Value
	Control	POC		
Overall	3.50	3.24	7.6 (0.4 to 14.3)	0.04
Diagnostic group				0.86*
ACS	3.43	3.15	8.2 (-0.9 to 16.5)	0.08
General	3.61	3.36	6.9 (-5.2 to 17.6)	0.25
Disposition				0.60*
Discharge home	3.68	3.50	4.9 (-5.9 to 14.5)	0.36
Admit to ward	3.66	3.22	12.1 (-0.8 to 23.3)	0.06
Admit to EMU	2.94	2.81	4.4 (-11.1 to 17.8)	0.56
Enrolling staff				0.06*
Consultant or registrar	3.63	2.94	19.1 (7.3 to 29.4)	<0.01
Junior medical officer	3.54	3.51	0.9 (-9.6 to 10.4)	0.85
Nurse	3.24	3.12	3.8 (-15.7 to 20.1)	0.68

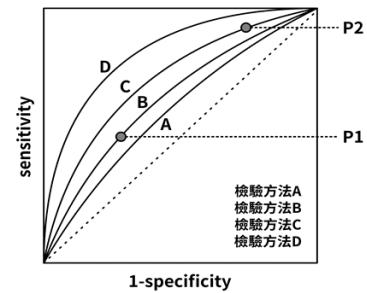
The bold text represent the results for the whole group.
 *Testing for an interaction between intervention and subgroup.
 ACS, acute coronary syndrome; EMU, Emergency Medicine Unit; POC, point-of-care.

- (A).POCT may shorten the disposition time overall.
 (B).Senior staffs (consultant or registrar) may shorten disposition time by POCT.
 (C).Junior medical officer may not shorten disposition time by POCT.
 (D).Nurses may shorten disposition time by POCT according to this study.
5. Usually the promoter regions are composed by exons. (是非題) (09 移至 block1)
6. Capillary electrophoresis on fluorescence-tagged PCR products is frequently used to detect gene mutation. This method is especially sensitive in detecting point mutation. (是非題) (09 移至 block1)
7. T (9;22) (q34; q11) results in translocation between BCR and ABL1 gene and leads to pathophysiological fusion protein. (是非題) (09 移至 block1)

§ B05 第一次區段考

1.關於右圖，何者正確？

- (1) 此圖為 Kaplan-Meier curve，適合評估檢驗對疾病的檢測效力
 - (2) 檢驗方法 D 在四種檢驗方法中，檢驗效力是最好的
 - (3) 檢驗方法 C 若以 P2 作為閾值（cut-off），可作為 screening test
 - (4) 檢驗方法 B 若以 P1 作為閾值，則須考量偽陽性的問題
 - (5) 檢驗方法 B 若以 P1 作為閾值，則有較高的陰性預測率（NPV）
- (A).123
(B).145
(C).234
(D).245
(E).345



2.About point of care testing (POCT), which is correct?

- (1) Shorten the therapeutic turnaround time due to faster laboratory tests,
 - (2) more expensive than those of central laboratory testing,
 - (3) reduces length of hospital stay due to faster laboratory services, and
 - (4) only can be applied within hospitals.
- (A).14
(B).123
(C).234
(D).1234

3. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in Junior emergency medicine unit (as below). According to the results, which of the following are correct?

- (1) POCT may shorten the disposition time overall.
- (2) Senior staffs (consultant or registrar) may shorten disposition time by POCT.
- (3) Junior medical officer may not shorten disposition time by POCT.
- (4) Nurses may shorten disposition time by POCT according to this study.

- (A).124
(B).123
(C).234
(D).134

	Geometric mean (hours)		% Reduction (95% CI)	p Value
	Control	POC		
Overall	3.50	3.24	7.6 (0.4 to 14.3)	0.04
Diagnostic group				0.86*
ACS	3.43	3.15	8.2 (−0.9 to 16.5)	0.08
General	3.61	3.36	6.9 (−5.2 to 17.6)	0.25
Disposition				0.60*
Discharge home	3.68	3.50	4.9 (−5.9 to 14.5)	0.36
Admit to ward	3.66	3.22	12.1 (−0.8 to 23.3)	0.06
Admit to EMU	2.94	2.81	4.4 (−11.1 to 17.8)	0.56
Enrolling staff				0.06*
Consultant or registrar	3.63	2.94	19.1 (7.3 to 29.4)	<0.01
Junior medical officer	3.54	3.51	0.9 (−9.6 to 10.4)	0.85
Nurse	3.24	3.12	3.8 (−15.7 to 20.1)	0.68

The bold text represent the results for the whole group.
*Testing for an interaction between intervention and subgroup.
ACS, acute coronary syndrome; EMU, Emergency Medicine Unit; POC, point-of-care.

4. Pharmacokinetics includes:

- (1) Drug absorption
- (2) Drug distribution
- (3) Drug-drug interaction in metabolism
- (4) Drug adverse effect
- (5) Drug excretion
- (6) Drug action at target site

Which is **CORRECT**?

- (A). 123456
- (B). 346
- (C). 124
- (D). 1236
- (E). 1235

5. Clinically utility of therapeutic drug monitoring (TDM) includes:

- (1) Compliance of patients with epilepsy
- (2) Patients with epilepsy in pregnancy
- (3) Adjust the dosage of tacrolimus in patients' status post renal transplantation
- (4) Adjust the dosage of amikacin in patients with septicemia
- (5) Adjust the dosage of rituximab in patients with diffuse large B cell lymphoma

Which ones are correct?

- (A). 12345
- (B). 1234
- (C). 123
- (D). 345

6. Which one of the following statements is wrong? (09 移至 block1)

- (A). The start codon is not necessarily in exon 1.
- (B). The stop codon is not necessarily in the last exon of that specific gene.
- (C). Untranslated regions are in exons.
- (D). Promotor regions are in introns.
- (E). mRNA sequences are the same as the coding sequences.

7. Which one is not a stop codon? (09 移至 block1)

- (A). TAA
- (B). TAG
- (C). TGA
- (D). TGG

8. FLT3 gene encodes a protein FLT3, which is important for hematopoietic cell development.

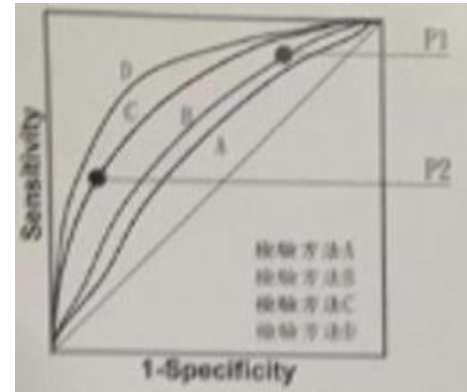
One type of mutation of this gene, FLT3-ITD (internal tandem duplication), which occurs exclusively in exon 14 and exon 15, is an insertional mutation. Interestingly, the numbers of insertional nucleotides are always the replicates of 3, and never disrupt the reading frame. From the above facts, which type of mutation is FLT3-ITD likely? (09 移至 block1)

- (A). Gain of function
- (B). Loss of function
- (C). Mis-sense mutation
- (D). Haploinsufficient
- (E). Non-sense mutation

§ B06 第一次區段考

1. 下圖是四種檢驗方法對於某疾病診斷能力的評估結果：

- (1) 這是 Receive operating characteristics (ROC) curve，用於評估檢驗對於疾病的診斷能力；
- (2) 檢驗方法 C 若以 P2 作為閾值（cut-on），則有較佳的陽性預測值（positive predictive value）；
- (3) 檢驗方法 A 和其它方法相比，其檢驗效力最好；
- (4) 檢驗方法 B 若以 P1 作為閾值，適合作為 screening test；
- (5) 檢驗方法 C 若以 P2 作為閾值，有較佳的陰性預測值（negative predictive value）



請問以上敘述何者正確？

- (A).124
- (B).135
- (C).234
- (D).145
- (E).345

（以下為第 2-4 題之案例）

王小明是 60 歲男性患者，除了肥胖的問題外，過去有糖尿病和高血壓病史，但是都沒有……，也沒有藥物控制。幾周前膝蓋疼痛，所以到西藥房買了止痛……。過了幾天，發現尿量減少，並有下肢水腫的情況。今日……全身倦怠，並有噁心、頭暈、呼吸急促、……心悸的症狀，因此到急診室，請回答以下問題：

2. 醫師幫王小明抽血檢查，同時進行尿液的檢測，得到初步臨床檢驗的結果如下：

- (1) 血中白血球計數 5000 /ul
- (2) 血鉀 X.5 mmol/L
- (3) 心電圖上……
- (4) C 型肝炎檢測為陰性
- (5) ……AST……3%

請問以上哪些是定量檢驗？

- (A).12
- (B).24
- (C).15
- (D).25
- (E).125

3. 承上題，檢驗人員打電話通報患者檢驗.....溶血.....的情況，可能造成血鉀濃度.....偏高，建議.....抽血檢測。請問，溶血對於血鉀檢測所造成的誤差，就試驗流程而言，主要是發生.....哪一階段？
- (A). 檢驗前.....檢驗造成
 - (B). 檢驗中.....檢驗測量所干擾
 - (C). 檢驗後.....發現問題造成
 - (D). (B)、(C)皆對
 - (E). (A)、(B)、(C)皆對
4. 承上題，重新抽血檢驗後，檢體已無溶血情況，血鉀濃度為 6 mmol/L，所以有高血鉀的情況。急診室醫師給予王小明初步的治療，並於 6 小時後再次抽血追蹤血中鉀離子濃度。已知血鉀檢驗的系統性誤差為 0 mmol/L，隨機誤差為 0.06 mmol/L，而再次檢驗的結果如下：
- (1) 6.1 mM
 - (2) 6.5 mM
 - (3) 5.0 mM
 - (4) 5.3 mM
 - (5) 5.9 mM
- 請問上述結果哪些代表測量值有顯著的變化？
- (A). 123
 - (B). 145
 - (C). 234
 - (D). 245
 - (E). 345
5. About point of care testing (POCT), which is **CORRECT**?
- (1) Reduces length of hospital stay due to faster laboratory services;
 - (2) more expensive than those of central laboratory testing;
 - (3) shorten the therapeutic turnaround time of laboratory tests; and
 - (4) only can be applied within hospitals.
- (A). 14
 - (B). 13
 - (C). 234
 - (D). 1234

6. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in emergency medicine unit (as below). According to the results, which one is **CORRECT**?

- (1) POCT may shorten the disposition time overall;
 - (2) Senior staffs (consultant or registrar) may shorten disposition time by POCT;
 - (3) Junior medical officer may shorten disposition time by POCT;
 - (4) Nurses may shorten disposition time by POCT according to this study.
- (A).24
(B).12
(C).23
(D).14

Table 2 Time from arrival to disposition decision

	Geometric mean (hours)		% Reduction (95% CI)	p Value
	Control	POC		
Overall	3.50	3.24	7.6 (0.4 to 14.3)	0.04
Diagnostic group				0.86*
ACS	3.43	3.15	8.2 (-0.9 to 16.5)	0.08
General	3.61	3.36	6.9 (-5.2 to 17.6)	0.25
Disposition				0.60*
Discharge home	3.68	3.50	4.9 (-5.9 to 14.5)	0.36
Admit to ward	3.66	3.22	12.1 (-0.8 to 23.3)	0.06
Admit to EMU	2.94	2.81	4.4 (-11.1 to 17.8)	0.56
Enrolling staff				0.06*
Consultant or registrar	3.63	2.94	19.1 (7.3 to 29.4)	<0.01
Junior medical officer	3.54	3.51	0.9 (-9.6 to 10.4)	0.85
Nurse	3.24	3.12	3.8 (-15.7 to 20.1)	0.68

The bold text represent the results for the whole group.
*Testing for an interaction between intervention and subgroup.
ACS, acute coronary syndrome; EMU, Emergency Medicine Unit; POC, point-of-care.

7. Pharmacokinetics includes:

- (1) Drug absorption
- (2) Drug distribution
- (3) Drug-drug interaction in metabolism
- (4) Drug action at target site
- (5) Drug adverse effect
- (6) Drug excretion

Which is **CORRECT**?

- (A).123456
(B).145
(C).125
(D).1234
(E).126

8. Clinically utility of therapeutic drug monitoring (TDM)

- (1)of beta-blocker in patients with hypertension
- (2) Patients with epilepsy in pregnancy
- (3) Adjust the dosage of tacrolimus for patients status post renal transplantation
- (4) Adjust the dosage of.....in patients with COVID-19 infection
- (5) Co-administration of.....and tacrolimus.....of kidney transplant complication with CNS.....infection

Which is **CORRECT**?

- (A).12345
(B).4

- (C).235
- (D).345

9. Please choose the most appropriate answer. (09 移至 block1)

- (A). It is very sensitive, and can detect specific DNA message from a single cell.
- (B). It is very quantitative, and can reliably determine the amount of expression of specific genes.
- (C). It can be very specific, especially with sequence-specific fluorescent probes and primers.
- (D). It is efficient and only takes hours to finish the reaction.
- (E). Radioactivity contamination should be considered when performing Q-PCR.

10. Which statement is the least appropriate for the basic structure of genes? (09 移至 block1)

- (A). Exon is usually much shorter than intron.
- (B). Promoters mean proteins that bind to the upstream DNA sequences of genes.
- (C). In mammalian cells, the initial amino acid of a nascent translated protein is methionine.
- (D). In mammalian cells, there are 3 different stop codons.
- (E). The strand of DNA that is complementary to mRNA is anti-sense strand DNA.

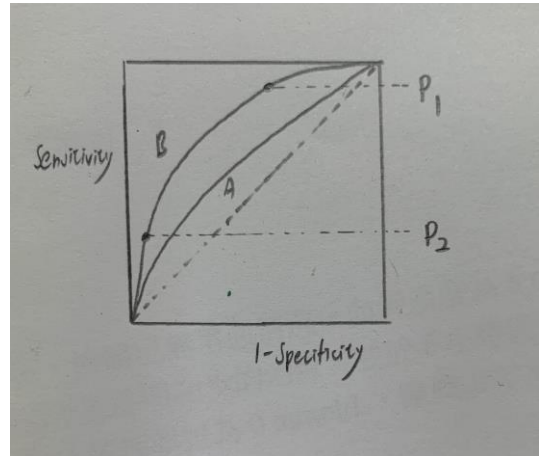
11. 解釋何為 CT? (問答題) (09 移至 block1)

12. 解釋何為 STR? (問答題) (09 移至 block1)

§ B07 第一次區段考

1. 下圖是兩種檢驗方法對某疾病診斷能力的結果？

- (1) 這是 Receive operating characteristics (ROC) curve，用於評估檢驗對於疾病的診斷能力
- (2) 檢驗方法 B 若以 P_2 作為閾值，則有較高的 LR^+
- (3) 檢驗方法 A 的檢驗效力較佳
- (4) 檢驗方法 B 若以 P_2 作為閾值，則有較高的 LR^-



請問以上敘述何者正確？

- (A).124
 - (B).134
 - (C).234
 - (D).123
2. 關於 covid-19 病毒抗原血清檢測，請問下列哪個族群 sensitivity 的估計值最低？
- (A).100 個高病毒量感染者與 100 個無感染者
 - (B).於每日有數萬例以上確認病例之社區，採檢有症狀者
 - (C).於台北市納入無症狀患者的族群
 - (D).ABC 差不多
 - (E).BC 差不多，但都比 A 低
3. 檢驗人員打電話來通知說病患的檢體呈現溶血 3+，可能會導致血鉀濃度偽性偏高，建議重新抽血取得檢體。請問溶血對血鉀濃度所造成的誤差，屬於檢驗流程中哪一階段的問題？
- (A).檢驗前 (Pre-analytic)：造成抽血過程的困難
 - (B).檢驗中 (Analytic)：檢驗方法學上面的干擾
 - (C).檢驗後 (Post-analytic)：報告結果發送上的問題
 - (D).AB 皆對
 - (E).ABC 皆對
4. 承上題，重新抽血檢驗後，檢體已無溶血情況，血鉀濃度為 6 mmol/L，所以有高血鉀的情況。急診室醫師給予王小明初步的治療，並於 6 小時後再次抽血追蹤血中鉀離子濃度。已知血鉀檢驗的系統性誤差為 0 mmol/L，隨機誤差為 0.06 mmol/L，而再次檢驗的結果如下：

- (1) 6.1 mM
- (2) 6.5 mM
- (3) 5.0 mM
- (4) 5.3 mM
- (5) 5.9 mM

請問上述結果哪些代表測量值有顯著的變化？

- (A).123
- (B).145
- (C).234
- (D).245
- (E).345

5. About point of care testing (POCT), which is **CORRECT**?

- (1) Reduces length of hospital stay due to faster laboratory services;
 - (2) more expensive than those of central laboratory testing;
 - (3) shorten the therapeutic turnaround time of laboratory tests; and
 - (4) only can be applied within hospitals.
- (A). 14
(B). 13
(C). 234
(D). 1234

6. There is a study about the impact of point-of-care-testing (POCT) on the disposition time (time from patient arrival to disposition decision) in emergency medicine unit (as below).

According to the results, which one is **CORRECT**?

- (1) POCT may shorten the disposition time overall;
- (2) Senior staffs (consultant or registrar) may shorten disposition time by POCT;
- (3) Junior medical officer may shorten disposition time by POCT;
- (4) Nurses may shorten disposition time by POCT according to this study.

- (A). 24
(B). 12
(C). 23
(D). 14

Table 2 Time from arrival to disposition decision

	Geometric mean (hours)		% Reduction (95% CI)	p Value
	Control	POC		
Overall	3.50	3.24	7.6 (0.4 to 14.3)	0.04
Diagnostic group				0.86*
ACS	3.43	3.15	8.2 (-0.9 to 16.5)	0.08
General	3.61	3.36	6.9 (-5.2 to 17.6)	0.25
Disposition				0.60*
Discharge home	3.68	3.50	4.9 (-5.9 to 14.5)	0.36
Admit to ward	3.66	3.22	12.1 (-0.8 to 23.3)	0.06
Admit to EMU	2.94	2.81	4.4 (-11.1 to 17.8)	0.56
Enrolling staff				0.06*
Consultant or registrar	3.63	2.94	19.1 (7.3 to 29.4)	<0.01
Junior medical officer	3.54	3.51	0.9 (-9.6 to 10.4)	0.85
Nurse	3.24	3.12	3.8 (-15.7 to 20.1)	0.68

The bold text represent the results for the whole group.

*Testing for an interaction between intervention and subgroup.

ACS, acute coronary syndrome; EMU, Emergency Medicine Unit; POC, point-of-care.

7. Pharmacokinetics includes:

- (1) Drug absorption
- (2) Drug distribution
- (3) Drug-drug interaction in metabolism
- (4) Drug action at target site
- (5) Drug adverse effect
- (6) Drug excretion

Which is **CORRECT**?

- (A). 123456
(B). 1256
(C). 1245
(D). 1234
(E). 1236

8. Clinically utility of therapeutic drug monitoring (TDM) includes: (多選)

- (A). Usage of ACEI on patients with heart failure.
- (B). Patients with epilepsy in pregnancy.
- (C). Adjust the dosage of amikacin in patients with septicemia.
- (D). Adjust the dosage of dexamethasone on COVID-19 patients.
- (E). Co-administration of voriconazole and tacrolimus on patients with kidney transplant and CNS fungal infection.

9. 有關胸腔 transudate 的特性，何者有誤？

- (A). 比重小於 1.015
(B). 白血球濃度低於 1000/mm³

- (C).取出的 protein 量小於 serum protein 的 50%
- (D).測得的 LDH 大於 60% serum 的 LDH
10. 兩檢體 A、B 皆放入 real time PCR 中測 mutant transcription 量。若已知檢體 A 的 mutant transcription 量是檢體 B 的十倍，且檢體 A 的 ct 值是 20.34，下列哪一個最有可能是檢體 B 的 ct 值? (09 移至 block1)
- (A).30.34
- (B).10.34
- (C).23.66
- (D).25.68
- (E).15.77
11. Which kind of mutation can be easily discerned by fluorescent capillary electrophoresis method? (09 移至 block1)
- (A).insertion or deletion
- (B).missense mutation
- (C).nonsense mutation
- (D).point mutation in well conserved region
- (E). insertion plus deletion with total nucleotide number remains unchanged
12. About quantitative real-time PCR which is inappropriate? (09 移至 block1)
- (A).It is very sensitive, and can detect specific DNA message from a single cell.
- (B).It is very quantitative, and can reliably determine the amount of expression of specific genes.
- (C).It can be very specific, especially with sequence-specific fluorescent probes and primers.
- (D).It is efficient and only takes hours to finish the reaction.
- (E). Radioactivity contamination should be considered when performing Q-PCR.

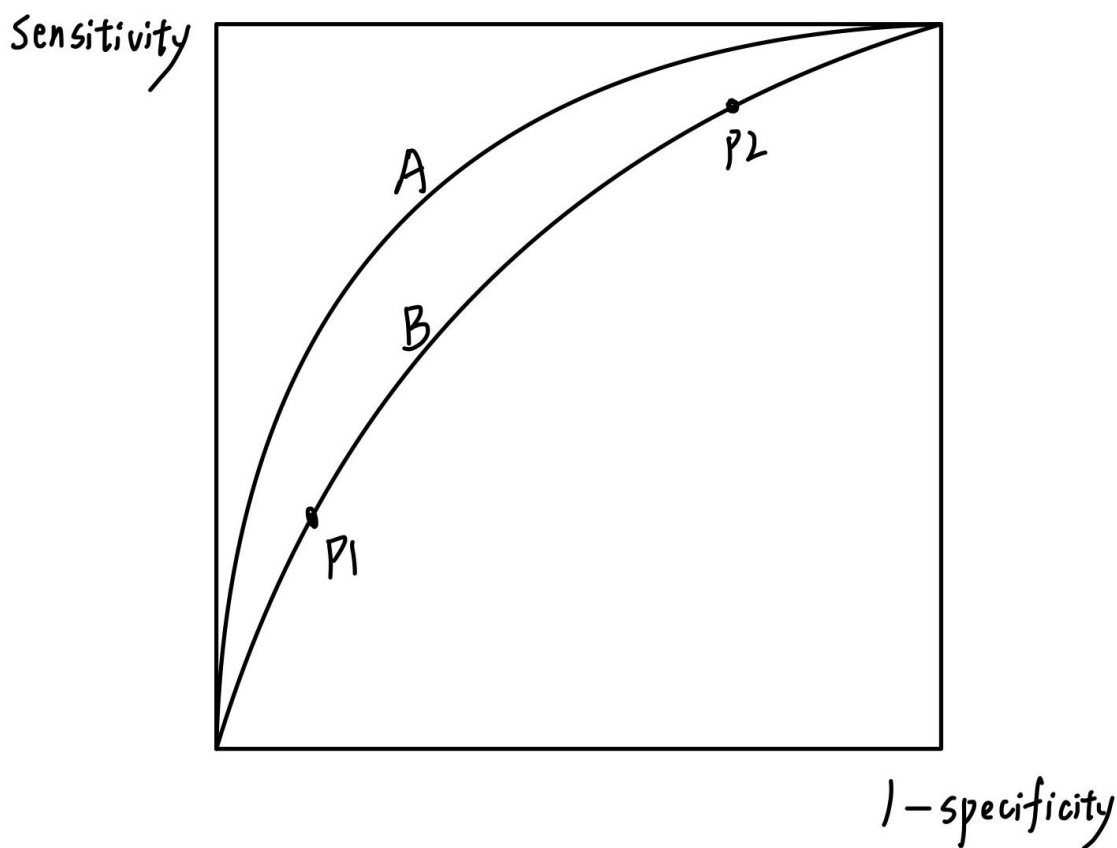
§ B08 第一次區段考

1. 下列何者是現行常見的 TDM ?
 - (A).EIS
 - (B).CLIA
 - (C).ELISA
 - (D).HPLC
 - (E). MS/MS
2. 下列何種病變會造成濾出液 ?
 - (A).胰臟炎
 - (B).腫瘤
 - (C).SLE
 - (D).心臟衰竭
3. 下列檢體何者不適用於 POCT ?
 - (A).血清
 - (B).唾液
 - (C).尿液
 - (D).全血
 - (E).鼻咽拭子
4. 下列何者應考慮使用 POCT ?
 - (A).流感重症患者
 - (B).HIV 患者
 - (C).伊波重症患者
 - (D).COVID 重症患者
 - (E). 以上皆是
5. TDM 相關敘述，何者正確 ?
 - (A).Phenobarbital、rifampin 是 CYP450 inducer，同時給其他藥會造成其他藥物濃度變低
 - (B).time-dependent 療效的藥，宜多次給藥
 - (C).concentration-dependent 的藥，懷疑有毒性，測 trough 濃度
 - (D).therapeutic index 小，即 therapeutic range 小的藥，更需要用 TDM 監測
 - (E). 以上皆是
6. 下列檢驗方法相關敘述何者正確 ?
 - (A).檢驗方法所造成的誤差只會發生在檢驗中，檢驗前和檢驗後皆不會發生
 - (B).可以透過重複檢驗標準差，決定 total analytic error
 - (C).不同檢驗平台，相同檢體所造成的誤差即為隨機性誤差(Random error)
 - (D).不同檢驗平台所造成的系統性誤差(Systemic error)……
 - (E).血鉀之最大可容許誤差(Total allowable error)可以訂為 20%
7. 下列何項檢體不是重點照護檢測 POCT 所用 ?
 - (A).血清
 - (B).尿液
 - (C).痰液
 - (D).全血
 - (E). 以上皆是
8. 檢查血清 serum 與腹水 albumin 濃度差異的目的為何 ?
 - (A).判斷是否正確採檢
 - (B).判斷檢體是否新鮮

- (C).判斷病人是否空腹
(D).判斷腹水為濾出液或滲出液
9. 在 TDM (therapeutic drug monitoring) 中，需要考慮藥物動力學 (pharmacokinetics) 的因素，主要包含下列哪幾項？
- (1) Drug absorption
 - (2) Drug distribution
 - (3) Drug metabolism
 - (4) Drug adverse effect
 - (5) Drug excretion
- (A).1, 2, 3, 4
 - (B).1, 2, 3, 5
 - (C).1, 2, 4, 5
 - (D).1, 3, 4, 5
 - (E).2, 3, 4, 5
10. 王小明是 60 歲男性患者，除了肥胖的問題外，過去有糖尿病和高血壓病史，但是都沒有……，也沒有藥物控制。幾周前膝蓋疼痛，所以到西藥房買了止痛……。過了幾天，發現尿量減少，並有下肢水腫的情況。今日……全身倦怠，並有噁心、頭暈、呼吸急促、……心悸的症狀，因此到急診室。重新抽血檢驗後，檢體已無溶血情況，血鉀濃度為 6 mmol/L，所以有高血鉀的情況。急診室醫師給予王小明初步的治療，並於 6 小時後再次抽血追蹤血中鉀離子濃度。已知血鉀檢驗的系統性誤差為 0 mmol/L，隨機誤差為 0.06 mmol/L，而再次檢驗的結果如下：
- (1) 6.1 mM
 - (2) 6.5 mM
 - (3) 5.0 mM
 - (4) 5.3 mM
 - (5) 5.9 mM
- 請問上述結果哪些代表測量值有顯著的變化？
- (A).123
 - (B).145
 - (C).234
 - (D).245
 - (E).345
11. 下列關於抽胸水檢查的敘述何者正確？
- (A).如果 eosinophil 含量高，代表有乳糜胸的可能
 - (B).如果中皮細胞含量高，代表有肺癌的可能
 - (C).好像跟腎有關……
 - (D).pH<6.5，代表食道破裂
12. (簡答) Please briefly explain why NGS can detect low-copy variants in a more unbiased way than conventional PCR?

§ B09 第一次區段考

1. (是非題) Fluorescence capillary electrophoresis is not appropriate for detecting point mutation.
2. (是非題) The lower the Ct number implies the higher amount of desired gene in the sample.
3. (問答題) 如果 mutated gene 佔 wild type 的部分之比例很少，如只佔 5%，則 PCR 就很難很精準的去找到相關的序列，而 NGS 技術卻可以做到這點，請簡述 how and why NGS 可以在這樣的檢測下有較高的靈敏度？（原文是英文內容，內容很接近 B08 檢醫考古 12 題）
4. (是非題) Real-time PCR can quantitate both DNA and RNA
5. (是非題) 5'untranslated regions are part of exon of a gene.
6. 請問下列敘述何者正確？



- (1) 此圖為 ROC curve，用以評斷檢驗方法的好壞
 - (2) P1 的 LR- 比 P2 的 LR- 還大
 - (3) P1 的 LR+ 比 P2 的 LR+ 還小
 - (4) 檢驗方法 B 比較好
- (A).12
(B).24
(C).14
(D).123
(E).234
7. 下列檢驗方法相關敘述何者正確？
- (A). 檢驗方法所造成的誤差只會發生在檢驗中，檢驗前和檢驗後皆不會發生
(B). 可以透過重複檢驗標準差，決定 total analytic error
(C). 不同檢驗平台，相同檢體所造成的誤差即為隨機性誤差(Random error)

(D).不同檢驗平台所造成的系統性誤差(Systemic error).....

(E).血鉀之最大可容許誤差(Total allowable error)可以訂為 20%

- 8.王小明是 60 歲男性患者，除了肥胖的問題外，過去有糖尿病和高血壓病史，但是都沒有.....，也沒有藥物控制。幾周前膝蓋疼痛，所以到西藥房買了止痛.....。過了幾天，發現尿量減少，並有下肢水腫的情況。今日.....全身倦怠，並有噁心、頭暈、呼吸急促、.....心悸的症狀，因此到急診室。重新抽血檢驗後，檢體已無溶血情況，血鉀濃度為 6 mmol/L，所以有高血鉀的情況。急診室醫師給予王小明初步的治療，並於 6 小時後再次抽血追蹤血中鉀離子濃度。已知血鉀檢驗的系統性誤差為 0 mmol/L，隨機誤差為 0.06 mmol/L，而再次檢驗的結果如下：

(1) 6.1 mM

(2) 6.5 mM

(3) 5.0 mM

(4) 5.3 mM

(5) 5.9 mM

請問上述結果哪些代表測量值有顯著的變化？

(A).123

(B).145

(C).234

(D).245

(E).345

- 9.下列檢體何者不適用於 POCT？

(A).血清

(B).唾液

(C).尿液

(D).全血

(E).鼻咽拭子

10. 下列何者可以作為 POCT 的對象？

(A).SARS-COVID-2

(B).血糖

(C).心臟節律

(D).hCG

(E).以上皆是

11. 下列何者應考慮使用 POCT？

(A).流感重症患者

(B).HIV 患者

(C).伊波重症患者

(D).COVID 重症患者

(E).以上皆是

12. 下列關於重點照護檢測的敘述，何者正確？

(A).優點是價格較便宜

(B).醫院並無重點照護檢測的需求，只需檢驗醫學部運作即可

(C).重點照護檢測的準確度都很高，可以依其進行臨床決策

(D).一般民眾也可以進行重點照護檢測，不需經由醫護人員

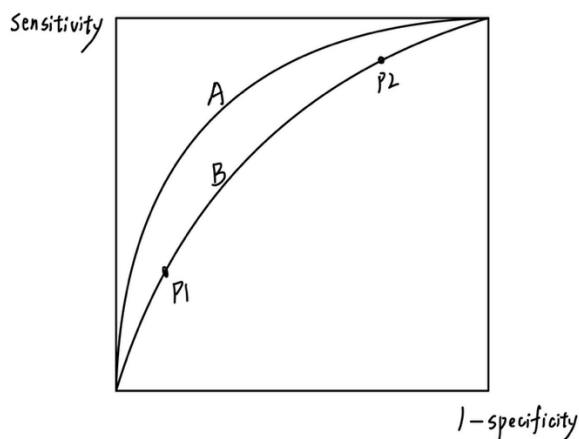
13. 檢測結果判讀須經醫事人員確認在 TDM (therapeutic drug monitoring) 中，需要考慮藥物動力學 (pharmacokinetics) 的因素，主要包含下列哪幾項？

(1) Drug absorption

- (2) Drug distribution
 - (3) Drug metabolism
 - (4) Drug adverse effect
 - (5) Drug excretion
 - (A).1, 2, 3, 4
 - (B).1, 2, 3, 5
 - (C).1, 2, 4, 5
 - (D).1, 3, 4, 5
 - (E).2, 3, 4, 5
14. TDM 可以檢測藥物濃度，請問下列敘述何者不適用？
- (A).2 個月大的男嬰服用 gentamicin 超過一週，醫師評估仍要繼續服用
 - (B).3 歲男童因氣喘發作
 - (C).6 歲女童癲癇，注射 methylprednisolone 及口服 prednisolone
 - (D).腎臟移植，tacrolimus
 - (E).56 歲男性曾因急性骨髓性白血病接受化療，因消化道潰瘍長期服用 esomeprazole。此次因為肺部 Aspergillosis，服用 voriconazole。
15. 下列何者是現行常見的 TDM？
- (A).EIS
 - (B).CLIA
 - (C).ELISA
 - (D).HPLC
 - (E).MS/MS
16. TDM 相關敘述，何者正確？
- (A).Phenobarbital、rifampin 是 CYP450 inducer，同時給其他藥會造成其他藥物濃度變低
 - (B).time-dependent 療效的藥，宜多次給藥
 - (C).concentration-dependent 的藥，懷疑有毒性，測 trough 濃度
 - (D).therapeutic index 小，即 therapeutic range 小的藥，更需要用 TDM 監測
 - (E). 以上皆是
17. 請問一下何者和 pleural effusion 無關
- (A).淋巴液回收
 - (B).血管通透性
 - (C).血球數量
 - (D).血液靜水壓
18. 檢查血清 serum 與腹水 albumin 濃度差異的目的為何？
- (A).判斷是否正確採檢
 - (B).判斷檢體是否新鮮
 - (C).判斷病人是否空腹
 - (D).判斷腹水為濾出液或滲出液
19. (簡答) Please briefly explain why NGS can detect low-copy variants in a more unbiased way than conventional PCR?

§ B10 第一次區段考

1. 在費城染色體 (Philadelphia chromosome) 相關的易位事件中，ABL1 與 BCR 兩基因分別位於哪兩條染色體上？
2. In real-time PCR, two samples A and B have Ct values of 25 and 30, respectively. What is the ratio of the amount of the target gene in samples A and B (A:B)?
3. 下列關於抽胸水檢查的敘述何者正確？
 - (A). 如果 eosinophil 含量高，代表有乳糜胸的可能
 - (B). 如果中皮細胞含量高，代表有肺癌的可能
 - (C). 好像跟腎有關.....
 - (D). pH<6.5，代表食道破裂
4. 請問下列關於此圖之敘述哪些正確？



- (1) 此圖為 ROC curve ,用以評斷檢驗方法的好壞
 - (2) P1 處的 negative predictive ratio 大於 P2
 - (3) P1 處的 LR+大於 P2
 - (4) B 相較於 A 是個比較好的檢定方法
- (A). 234
 - (B). 12
 - (C). 123
 - (D). 14
5. 下列檢驗方法相關敘述何者正確？
 - (A). 檢驗方法所造成的誤差只會發生在檢驗中，檢驗前和檢驗後皆不會發生
 - (B). 可以透過重複檢驗標準差，決定 total analytic error
 - (C). 不同檢驗平台，相同檢體所造成的誤差即為隨機性誤差(Random error)
 - (D). 不同檢驗平台所造成的系統性誤差(Systemic error).....
 - (E). 血鉀之最大可容許誤差(Total allowable error)可以訂為 20%
 6. 檢驗人員打電話來通知說病患的檢體呈現溶血 3+，可能會導致血鉀濃度偽性偏高，建議重新抽血取得檢體。請問溶血對血鉀濃度所造成的誤差，屬於檢驗流程中哪一階段的問題？
 - (A). 檢驗前 (Pre-analytic)：造成抽血過程的困難
 - (B). 檢驗中 (Analytic)：檢驗方法學上面的干擾
 - (C). 檢驗後 (Post-analytic)：報告結果發送上的問題
 - (D). AB 皆對
 - (E). ABC 皆對

-
7. 承上題，重新抽血檢驗後，檢體已無溶血情況，血鉀濃度為 6 mmol/L，所以有高血鉀的情況。急診室醫師給予王小明初步的治療，並於 6 小時後再次抽血追蹤血中鉀離子濃度。已知血鉀檢驗的系統性誤差為 0 mmol/L，隨機誤差為 0.06 mmol/L，而再次檢驗的結果如下：

- (1) 6.1 mM
- (2) 6.5 mM
- (3) 5.0 mM
- (4) 5.3 mM
- (5) 5.9 mM

請問上述結果哪些代表測量值有顯著的變化？

- (A). 145
 - (B). 234
 - (C). 245
 - (D). 345
-

- 11 下列檢體何者不適用於 POCT？

- (A). 血清
 - (B). 唾液
 - (C). 尿液
 - (D). 全血
 - (E). 鼻咽拭子
-

- 12 下列何者可以作為 POCT 的對象？

- (A). SARS-COVID-2
 - (B). 血糖
 - (C). 心臟節律
 - (D). hCG
 - (E). 以上皆是
-

- 13 哪些問題必須使用 POCT？

- (A). 流感重症患者
 - (B). HIV 患者
 - (C). 伊波拉感染者
 - (D). COVID19 重症
 - (E). 以上皆是
-

- 14 下列關於重點照護檢測的敘述，何者正確？

- (A). 重點照護檢測的優點是價格便宜
 - (B). 醫院並無重點照護檢測的需求，只需檢驗醫學部運作即可
 - (C). 重點照護檢測的準確度皆很高，可以依其進行臨床決策
 - (D). 一般民眾也可以進行重點照護檢測，不需經由醫護或醫事人員
 - (E). 重點照護檢測的判讀需經醫事人員確認
-

-
- 15 檢測結果判讀須經醫事人員確認在 TDM (therapeutic drug monitoring) 中，需要考慮藥物動力學 (pharmacokinetics) 的因素，主要包含下列哪幾項？
- (1) Drug absorption
 - (2) Drug distribution
 - (3) Drug metabolism
 - (4) Drug adverse effect
 - (5) Drug excretion
- (A). 1, 2, 3, 4
(B). 1, 2, 3, 5
(C). 1, 2, 4, 5
(D). 1, 3, 4, 5
(E). 2, 3, 4, 5
-
- 16 TDM 可以檢測藥物濃度，請問下列敘述何者不適用？
- (A). 2 個月大的男嬰服用 gentamicin 超過一週，醫師評估仍要繼續服用
 - (B). 3 歲男童因氣喘發作
 - (C). 6 歲女童癲癇，注射 methylprednisolone 及口服 prednisolone
 - (D). 腎臟移植，tacrolimus
 - (E). 56 歲男性曾因急性骨髓性白血病接受化療，因消化道潰瘍長期服用 Esomeprazole。此次因為肺部 Aspergillosis，服用 voriconazole。
-
- 17 下列何者不是 TDM 常見的檢驗方法？
- (A). Electrochemical Impedance Spectroscopy (EIS)
 - (B). Chemiluminescent Immunoassay (CLIA)
 - (C). Enzyme-linked Immunosorbent Assay (ELISA)
 - (D). High Performance Liquid Chromatography (HPLC)
 - (E). Tandem Mass spectrometry (MS/MS)
-
- 18 有關 TDM 的相關敘述，何者正確？
- (A). Phenobarbital, Rifampin 是 CYP450 inducers，同時給其他藥會造成其他藥物濃度變低。
 - (B). Time-dependent 療效的藥，宜多次給藥。
 - (C). Concentration-dependent 的藥，懷疑有毒性，測 Trough 濃度。
 - (D). Therapeutic index 小，即 Therapeutic range 小的藥，更需要用 TDM 檢測。
 - (E). 以上皆是
-
- 19 What is not related to pleural effusion?
- (A). Osmotic pressure of blood vessels
 - (B). Permeability of blood vessels
 - (C). Hydrostatic pressure of blood vessels
 - (D). Number of blood cells in blood vessels
-
- 20 檢查血清 (serum) 與腹水蛋白質濃度差異之目的為何？
- (A). 判斷是否正確採檢
 - (B). 判斷檢體是否新鮮
 - (C). 判斷病人是否空腹
 - (D). 判斷腹水為濾出液或滲出液
-
- 21 關於體腔液檢測何者錯誤
- (A). 滲出液(exudate)常出現於癌症轉移
 - (B). 腹腔灌洗(peritoneal washing)是由手術中取得，可知癌症是否轉移
 - (C). 胸水的酸鹼值和澱粉酶是食道破裂的指標
 - (D). 乳糜胸是腎病症候群
-

-
- 22 關於胸腔滲出液的特性，下列何者錯誤？
- (A). 如未加入抗凝劑，則過一段時間後會凝固
 - (B). 測得的 Total Protein 大於血清 Total Protein 的 50%
 - (C). 測得的 Lactate Dehydrogenase (LDH) 小於血清 LDH 的 50%
 - (D). 比重大於 1.018
-
- 23 STR is a marker used for identification of individual in forensic medicine. Please write the full name of STR.
-
- 24 (是非) 5-UTR is composed by intron.
-
- 25 下列何者是現行常見的 TDM？
- (A). EIS
 - (B). CLIA
 - (C). ELISA
 - (D). HPLC
 - (E). MS/MS
-